

Josh Steckel

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EDUCATION

University of Delaware

Expected May 2029

Bachelor of Engineering in Electrical Engineering | GPA: 3.99

EXPERIENCE

Ground Station Team Lead

May 2026 – Present

Delaware Atmospheric Plasma Probe Experiment (DAPPER)

- Led a 5-member subteam designing and upgrading ground station support equipment for a student-built Low-Earth Orbit (LEO) CubeSat mission
- Developed and tested channel models in MATLAB to determine optimal constant and adaptive downlink data rates for the Ultra-High Frequency (UHF), Audio Frequency Shift Keying (AFSK) spacecraft communications link, achieving 359 kB/day data transmission at 12 dB link margin
- Evaluated mission link budget and communications system performance for NASA's CubeSat Launch Initiative Pre-Procurement Readiness Review (PPRR) and FCC licensing, supporting technical documentation and regulatory compliance reviews

PROJECTS

Altoids Tin Meshtastic Node | PCB Design, KiCad, ESP32, LoRa, RF Design, C++

- Designed a low-cost 55 × 75 mm, 2-layer 915 MHz Meshtastic PCB for an Altoids tin, using CPWG impedance-controlled routing; integrated power charging/regulation and removable display interfaces, achieving 30-hour runtime and successful field messaging/display operation

University Transit System Departure Board | Embedded Systems, ESP32, FreeRTOS, AWS, Python, C++

- Designed a compact OLED arrival/departure board compatible with university transit systems, built on an automated headless web scraper (Selenium, Docker) hosted on AWS Lambda
- Implemented multi-core Real-Time Operating System (RTOS) firmware on a low-power ESP32 client fetching pre-processed JSON data over secure HTTPS, cutting load time by 75% while running a horizontally scrolling marquee display

Fault Injection on PQC Algorithm Research | Hardware Security, ChipWhisperer, ARM, Post-Quantum Cryptography

- Conduct fault-injection experiments with ChipWhisperer to induce instruction skips on ARM microcontrollers executing Kyber
- Analyze device responses to tune glitch parameters and identify reliable fault-injection windows

Power Side-Channel Attacks Research | Machine Learning, CNNs, Signal Processing, Python

- Trained and tested Convolutional Neural Network (CNN) models on 60+ hours of Virtual Reality (VR) device power consumption data to classify device activity from Short Time Fourier Transform (STFT) frequency-spectrum images, achieving 97% classification accuracy and strengthening applied signal processing and data analysis skills

Folded Dipole Design and Installation | Antenna Design, RF Engineering, Impedance Matching, NanoVNA

- Applied antenna theory to design and tune a 145 MHz four-element folded dipole antenna, replacing the Amateur Radio Club's damaged repeater antenna
- Optimized antenna gain and matched its impedance to the coaxial cable, recovering full repeater coverage

SKILLS

Core Skills: PCB Design, RF and Digital Signal Processing, Embedded Systems, FPGA Design, Systems Programming

Lab Equipment: Oscilloscope, Spectrum Analyzer, Vector Network Analyzer (VNA), Multimeter

Programming Languages: C, C++, Python, MATLAB, Verilog, VHDL

Software & Tools: KiCad, Vivado, GNU Radio, Git, SolidWorks

ACTIVITIES, LEADERSHIP & CERTIFICATIONS

- Amateur Radio Club — Treasurer
 - Manage project material costs and record purchases for reimbursement
- FCC Amateur Radio License — Technician Class (KD3DKR)